

Status, 24 Sep 2026. New bank: 50k $\theta \times 2$ patterns per family, $\bar{n} \in [75, 950]$, 12k test θ per family. $\tilde{\delta} = \text{ext}$ (1 = 5% rejection boundary). PersLay: classification only; PersLay-z at 2/10 seeds.

Classification — accuracy (overall) and recall (per family)

	all	Pois.	Thomas	nested	Matérn	LGCP
L, F, G, J	0.604	0.812	0.160	0.611	0.882	0.552
L (VIHRS)	0.583	0.817	0.150	0.571	0.890	0.487
PersLay-z DTM ₁₀ H_{0+1}	0.585	0.799	0.147	0.596	0.845	0.536
PersLay DTM ₁₀ H_{0+1}	0.577	0.830	0.130	0.600	0.853	0.470
PI alpha H_{0+1}	0.563	0.828	0.119	0.562	0.771	0.533
PI DTM ₁₀ H_{0+1}	0.549	0.802	0.144	0.585	0.674	0.539

Classification by $\tilde{\delta}$ — accuracy, Poisson excluded (chance 0.25)

$\tilde{\delta}$	0.05	0.10	0.18	0.35	0.66	1.26	2.41	4.61	8.79	16.8
L, F, G, J	.279	.338	.379	.485	.595	.626	.684	.773	.819	.831
PersLay-z DTM ₁₀ H_{0+1}	.256	.309	.332	.404	.517	.605	.672	.755	.806	.823
PI alpha H_{0+1}	.195	.227	.249	.304	.418	.563	.651	.739	.794	.817
PI DTM ₁₀ H_{0+1}	.201	.225	.234	.270	.335	.491	.652	.758	.809	.828

Parameters — log-RMSE, pooled over targets

	Poisson	Thomas	nested	Matérn	LGCP
L, F, G, J	0.067	0.754	0.571	0.111	0.535
L (VIHRS)	0.066	0.757	0.576	0.122	0.535
PI DTM ₅ H_{0+1}	0.067	0.777	0.595	0.151	0.563
PI alpha H_{0+1}	0.067	0.795	0.611	0.104	0.569
PI Rips H_0	0.067	0.846	0.648	0.103	0.595

Parameters by $\tilde{\delta}$ — paired log-RMSE, PH – L, F, G, J (+ = PH worse; * = 95% CI excludes 0)

	0.05	0.10	0.18	0.35	0.66	1.26	2.41	4.61	8.79	16.8
Matérn PI alpha H_{0+1}	+.004	+.002	+.004*	+.003*	.000	−.002*	−.010*	−.030*	−.029*	−.039*
Thomas PI DTM ₅ H_{0+1}	−.004	+.013	+.009	+.015*	+.041*	+.049*	+.052*	+.030*	+.054*	+.005
nested PI DTM ₅ H_{0+1}	+.005	+.010	+.003	+.004	+.007*	+.023*	+.037*	+.027*	+.037*	+.026*
LGCP PI DTM ₅ H_{0+1}	+.049*	+.050*	+.033*	+.050*	+.062*	+.054*	+.040*	+.016*	+.008*	−.017*

L, F, G, J = classical arm on the **fixed** grid. PI = persistence image. $\tilde{\delta} < 0.05$ (unbinned): Thomas 42%, LGCP 39%, nested 13%, Matérn 10%.